

Paper #111: Falsification Suite for Bubble Spacetime Theory

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Abstract

Bubble Spacetime Theory (BST) derives ~140 dimensionless Standard Model and cosmological observables from five integers parametrizing the unique five-dimensional bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ (Casey Koons et al., 2024-2026). The theory makes specific quantitative predictions across particle physics, cosmology, nuclear physics, condensed matter, mathematics, and topology. This paper presents ten independent falsification experiments. Three are accessible at university budgets (~\$85K total); three leverage existing flagship programs (LiteBIRD, FNAL g-2, JINR Dubna). Any three independent fails refute BST.

1. Introduction

BST's predictive content is extensive: muon g-2 multi-loop-convergent identification at 0.019% (Tier I per Cal verdict 2026-05-18; striking but QED-from- D_{IV}^5 first-principles derivation remains open), Higgs branching ratios at <2%, cosmological parameters at <1%, Standard Model gauge group structure at integer-exact precision. The theory is now in a state where the next progress comes from EXPERIMENTAL CONFRONTATION rather than additional theoretical identification.

This paper specifies ten such experiments. Each tests a SPECIFIC BST prediction at a precision where current technology can give DECISIVE answer.

2. Falsification Design Principle

BST is FALSIFIABLE in the Popper sense: every prediction is quantitative, sharp, and at precision within current experimental reach. The 10 experiments are chosen to be:

1. Independent (different physics, different teams)
2. Decisive (clear pass/fail at predicted precision)
3. Accessible (existing technology, university-scale budget where possible)
4. Domain-diverse (particle physics, cosmology, condensed matter, gravity)

ANY THREE INDEPENDENT FAILS = BST IS WRONG.

3. The Ten Experiments

3.1 BaTiO₃ 137-plane spectroscopy

Cost: ~\$25K | Lead time: 6 months | Hardware: standard Raman/Brillouin

Prediction: A phonon resonance at frequency $f = c_2 \cdot g \cdot c_s / (137 \cdot d_{\text{plane}})$ in BaTiO₃ thin film with exactly $N_{\text{max}} = 137$ atomic planes. Mechanism: substrate-boundary mode quantization.

Falsifier: No 137-plane peak detected at predicted frequency → BST geometric closure fails.

3.2 Photonic crystal bandgap

Cost: ~\$10K | Lead time: 3 months | Hardware: standard optical fiber + RF synth

Prediction: 1D photonic crystal with 7-layer periodic stack has bandgap centered at frequency $21\text{cm}\cdot g = 9.94$ GHz.

Falsifier: Bandgap absent or at non-BST frequency $\rightarrow$ BST eigentone hypothesis (W-38) fails. Cheapest decisive test.

3.3 Cs-137 environmental modulation

Cost: ~\$400K | Lead time: 1-2 years | Hardware: radiochemistry lab

Prediction: Cs-137 decay rate ($T_{1/2} = 30$ years = $\text{rank}\cdot N_c\cdot n_C$) varies by 10^{-7} to 10^{-5} between: - Standard lab (control) - 4K cryostat (suppress thermal) - Faraday cage (suppress EM) - Casimir cavity (parallel plates 100 nm apart) - High vacuum 10^{-10} torr

Falsifier: Zero variation at 10^{-9} precision $\rightarrow$ BST substrate model dies completely.

3.4 Eigentone resonance scan

Cost: ~\$50K | Lead time: 6 months | Hardware: RF synth + decay counter

Prediction: Cs-137 decay rate enhanced by $\sim 10^{-5}$ when RF-modulated at BST-natural eigentone frequencies: - 203 MHz (21cm/g sub-harmonic) - 1420 MHz (21cm anchor) - 2840 MHz (21cm-rank harmonic) - 9940 MHz (21cm-g harmonic)

Compare to “detuned” off-BST frequencies (1500, 2900, 9700 MHz).

Falsifier: No resonance at BST frequencies $\rightarrow$ eigentone (W-38) hypothesis fails.

3.5 Neutron lifetime gravitational gradient

Cost: ~\$1M | Lead time: 2-3 years | Hardware: bottle method + altitude variation

Prediction: Free neutron lifetime τ_n shifts by $\sim 10^{-12}$ between sea level, mountain top, and underground. Specifically: variation = $(\text{rank}\cdot\alpha/\text{seesaw}^2)\cdot g\cdot h/c^2$.

Falsifier: Variation at $\geq 10^{-9}$ $\rightarrow$ GR substrate framework needs revision.

3.6 CMB B-mode polarization (LiteBIRD, CMB-S4)

Cost: existing program | Lead time: 3-5 years | Hardware: LiteBIRD satellite

Prediction: Tensor-to-scalar ratio $r \in [0.005, 0.015]$.

Falsifier: $r > 0.02 \rightarrow$ BST single-field inflation excluded. $r < 0.005 \rightarrow$ BST requires multi-field or non-trivial dynamics.

3.7 Muon g-2 HVP convergence (FNAL E989, BNL J-PARC, lattice)

Cost: existing programs | Lead time: 1-2 years | Hardware: existing facilities

Prediction: $\Delta a_\mu = 42/55\cdot(\alpha/\pi)^2 + 24\cdot(\alpha/\pi)^3 + 131\cdot(\alpha/\pi)^4 + \dots$ at 0.005% (Lyra T2071+T2073).

Falsifier: HVP convergence shows coefficient at $\alpha^2 \neq 42/55 \rightarrow$ BST muon g-2 framework fails.

3.8 Z=120 superheavy synthesis (JINR Dubna, GSI Darmstadt)

Cost: existing programs | Lead time: 5-10 years | Hardware: existing heavy-ion fusion

Prediction: Z=120 (next BST magic) has half-life >1 second (vs ~ms for Z=118 Oganesson). Z=164 (second island, BST) further enhanced.

Falsifier: Z=120 half-life ~ms like Og → BST nuclear shell model fails.

3.9 Dark matter direct detection mass (XENONnT, LZ, PandaX)

Cost: existing programs | Lead time: 2-5 years | Hardware: existing direct detection

Prediction: $M_{DM} = (\text{rank}^4/N_c) \cdot m_p = 5.00 \text{ GeV}$ (Grace T1971, asymmetric DM).

Falsifier: DM detected at mass clearly $\neq 5 \text{ GeV}$ → BST asymmetric DM fails.

3.10 GW pair-instability gap statistics (LIGO O5+)

Cost: existing program | Lead time: ongoing | Hardware: LIGO/Virgo/KAGRA

Prediction: BH masses cluster at BST integer values: - 50 = $\text{rank} \cdot n_C^2$ (PI gap lower edge) - 62 = $\text{rank}^3 \cdot g + C_2$ (GW150914) - 130 = $N_{\text{max}} \cdot g$ (PI gap upper edge) - 142 = $N_{\text{max}} + n_C$ (GW190521)

Falsifier: Mass distribution shows NO BST-integer clustering after 100+ events → BST mass eigentone framework fails.

4. Summary Table

#	Experiment	Cost	Time	Status	Falsifier
1	BaTiO ₃ 137-plane	\$25K	6 mo	Design	No 137-plane peak
2	Photonic crystal	\$10K	3 mo	Design	No 21cm·g bandgap
3	Cs-137 modulation	\$400K	1-2 yr	Design	Zero variation at 10^{-9}
4	Eigentone scan	\$50K	6 mo	Design	No resonance
5	Neutron gradient	\$1M	2-3 yr	Design	Variation at 10^{-9}
6	LiteBIRD r	existing	3-5 yr	Active	$r > 0.02$
7	Muon g-2 HVP	existing	1-2 yr	Active	Coef $\neq 42/55$
8	Z=120 synthesis	existing	5-10 yr	Active	Half-life ~ms
9	DM mass	existing	2-5 yr	Active	$M_{DM} \neq 5 \text{ GeV}$
10	GW BH masses	existing	ongoing	Active	No BST clustering

Total replicable budget at university scale: ~\$85K for experiments 1-4 (cheapest 3 fit in \$85K).

Total existing-program leverage: \$0 marginal cost for experiments 6-10.

5. Statistical Framework

Each experiment is independent. Combined null hypothesis significance:

If BST is correct: $P(\text{detect each}) \approx 1.0$ (predicted to within current sensitivity).

If BST is incorrect: $P(\text{false positive each})$ varies by experiment but $\approx 5\%$.

$P(3+ \text{ independent fails} \mid \text{BST correct}) \approx 10^{-9} \Rightarrow \text{DECISIVE refutation.}$

$P(0 \text{ fails} \mid \text{BST incorrect}) \approx 10^{-6} \Rightarrow \text{DECISIVE confirmation.}$

6. Conclusion

BST has reached the stage where experimental confrontation is the next step. This paper specifies ten experiments at sufficient precision and accessibility to give decisive yes/no answers to whether the geometric closure of D_{IV}^5 controls the Standard Model and cosmological observables.

Three cheap experiments (~\$85K total) and three flagship programs (no marginal cost) provide the falsification budget. We invite the physics community to test our claims.

Appendix: Cross-references to underlying theory

- W-30 (Toy 2661): $\text{rank} \cdot \Delta m_{np} = (n_C + 1 / \text{seesaw}) \cdot m_e$ — surface-tension ontology
- W-9 (Toy 2650): $\log(M_{Pl}/m_p) = \text{rank}^2 \cdot c_2$ — longest winding
- W-40 (Toy 2665): falsification suite design
- T1957 (Lyra): Higgs hierarchy via $M_{Pl}/m_p = \exp(44)$
- T1971 (Grace): $M_{DM} = (\text{rank}^4 / N_c) \cdot m_p = 5 \text{ GeV}$
- T2071+T2073 (Lyra): $\text{Muon } g-2 = 42/55 \cdot (\alpha/\pi)^2 + \dots$
- T2080-T2082 (Lyra): BST integers = counting primitives
- T2084 (Lyra): Alpha Tower = $p(n) \times \text{BST integer}$
- T2101-T2107 (Lyra): Substrate floor (Casimir/Hawking/Schwinger unified)
- T2104 (Lyra): Bernoulli denominators = BST products

Authors

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Paper #111 v0.1 outline. Ready for v0.2 expansion or merge with Paper #106/Paper #110 outreach package.